Installing PostgreSQL & pgAdmin on your laptop

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This installation guide explains how to install PostgreSQL and pgAdmin on your laptop using Docker. PostgreSQL is a database server, and pgAdmin is a user interface that lets you run SQL queries.

This installation guide is intended for laptops running Windows 11 or a recent version of macOS.

Why you need to install PostgreSQL and pgAdmin

SQL is one of the topics covered in this course. To learn SQL properly, you need to practise using it. PostgreSQL and pgAdmin let you run SQL queries yourself and view the results.

Installing containers using Docker Desktop

In this course, we use Docker Desktop to run PostgreSQL and pgAdmin in containers on your laptop. This ensures that everyone uses the same software and configuration.

What is Docker Desktop?

Docker Desktop is an application that makes it easy to download, start, stop and manage containers on your laptop.

What is a container?

A container is an isolated environment that runs an application together with the software it needs to function. This means you do not need to install PostgreSQL and pgAdmin separately on your laptop.

Step 1 - Download and nstall Docker Desktop

If Docker Desktop is already installed on your computer, you can skip step 1.

You can download Docker Desktop from the following webpage:

https://www.docker.com/products/docker-desktop/

Click the ‘Download Docker Desktop’ button.

Different versions are available for different operating system / CPU combinations. Choose the version that matches your computer.

For Windows

1. Right-click the **Start button**.
2. Select **System**.
3. Find **System type** under *Device specifications*.
4. Check which processor architecture is listed.

What does it say?	Which Docker Desktop version should you download?
x64-based processor	Windows – AMD64
ARM-based processor	Windows – ARM64

Note: AMD64 is also the correct choice for computers with an Intel processor. The name AMD64 refers to the processor architecture, not the brand.

For macOS

1. Click the Apple logo (🍎) in the top-left corner.
2. Select About This Mac.
3. Check which processor or chip is listed.

What does it say?	Which Docker Desktop version should you download?
Intel	Mac – Intel Chip

What does it say?	Which Docker Desktop version should you download?
Apple M1, M2, M3, M4, M5 or newer	Mac – Apple Silicon

Install Docker Desktop

Once the download is complete, open the downloaded installation file.

- **Windows**: Run the installer and follow the on-screen instructions.
- **macOS**: Open the downloaded .dmg file and drag Docker to the Applications folder.

If prompted, restart your computer to complete the installation.

Step 2 - Check that Docker Desktop is working

Once Docker Desktop is installed, check that it runs without any error messages.

Start Docker Desktop

For Windows

- Click the Windows Start button.
- Type `Docker` in the search field.
- Click the Docker Desktop application.
- Wait until the application has fully started. This may take a few minutes the first time.

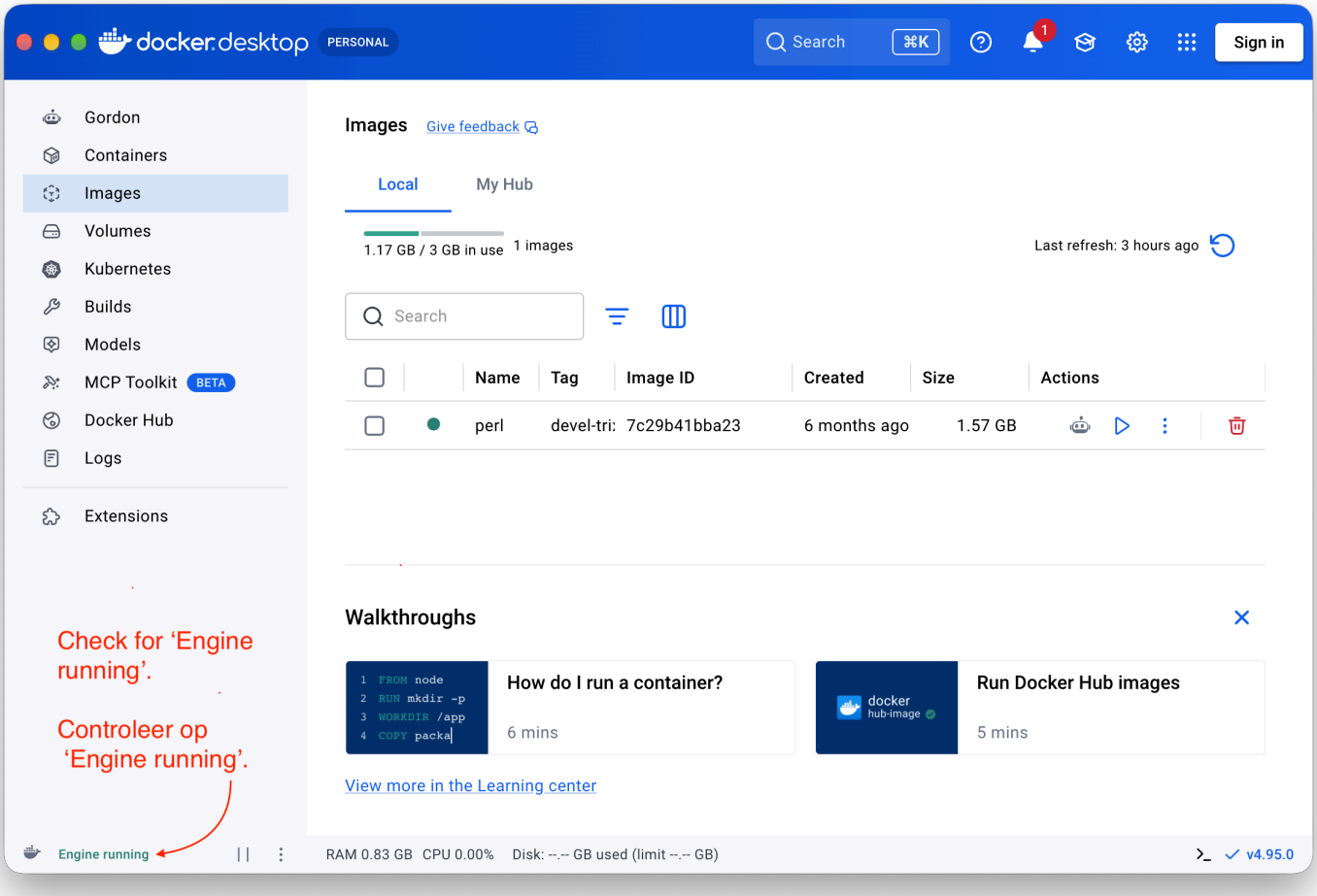
For macOS

- Open Docker Desktop using Spotlight (Command + Space, type `Docker`).
- Wait until Docker Desktop has fully started. This may take a few minutes the first time.

Check that the Docker Engine is running

For Windows and macOS

- Open Docker Desktop.
- Check the Docker Engine status in the bottom-left corner of the window.
- If it says Engine running, Docker is working correctly.
- If the Docker Engine is still starting, wait until it is running.



> Docker Desktop may look different from the screenshot. The main thing to check is that 'Engine running' appears in the bottom-left corner.

Only continue once Docker Desktop is running without any error messages.

Step 3 - Download the installation files

- Download the ZIP file containing the installation files using the URL below.

https://github.com/hogeschool/Data-Technologies-public/blob/main/Kickoff/data/files/data-technologies-pgsql.zip

By default, the file is saved in the `Downloads` folder.

For Windows

- Open File Explorer (Windows + E shortcut).
- Click `Downloads` on the left.
- Find the file `data-technologies-pgsql.zip`.
- Right-click the file.
- Select `Extract All....`
- Leave the suggested location unchanged.
- Click `Extract`.

Windows now creates the `data-technologies-pgsql` folder.

For macOS

- Open Finder.
- Click `Downloads`.
- Find the file `data-technologies-pgsql.zip`.
    - If you can already see the `data-technologies-pgsql` folder, the ZIP file has been extracted.
    - If only the ZIP file is present, double-click it to extract it.

The extracted files

After extraction, the `data-technologies` folder is in the `Downloads` folder and contains the following structure:

```
data-technologies/
└── docker-compose/
    └── postgresql/
```

```
        ├── compose.yaml
        ├── servers.json
        └── university.sql
```

On Windows, this folder may still be inside the extracted `data-technologies-pgsql` folder.

Step 4 - Move the files to Documents

Move the `data-technologies` folder to your personal `Documenten` or `Documents` folder.

The files will then be in the following location:

For macOS

```
~/Documents/data-technologies/docker-compose/postgresql
```

For Windows

```
Documents\data-technologies\docker-compose\postgresql
```

Important: Do not delete or move this folder after installing PostgreSQL. Docker uses the configuration files in this folder.

Step 5 - Install PostgreSQL and pgAdmin

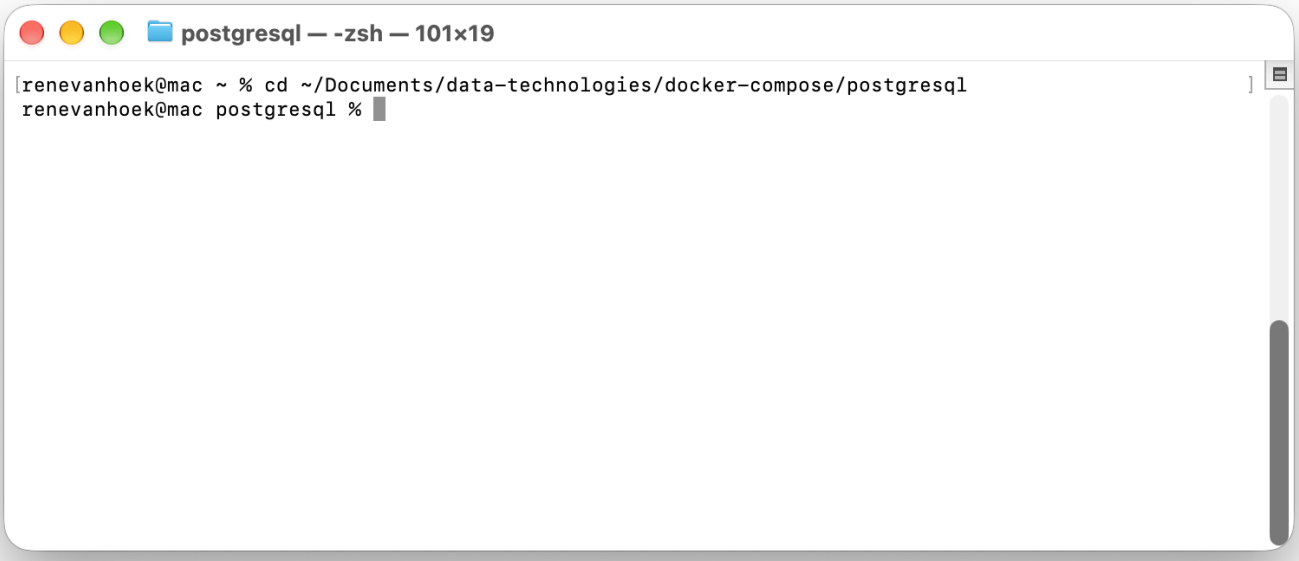
We will now install PostgreSQL and pgAdmin using Docker Compose.

Open the terminal

For macOS

- Open Terminal using Spotlight (Command + Space, type `Terminal`).
- Run the following command:

```
cd ~/Documents/data-technologies/docker-compose/postgresql
```



For Windows

- Open File Explorer.
- Navigate to `Documenten → data-technologies → docker-compose → postgresql`
- Click the File Explorer address bar.
- Type `powershell` and press Enter.

PowerShell now opens directly in the correct folder.

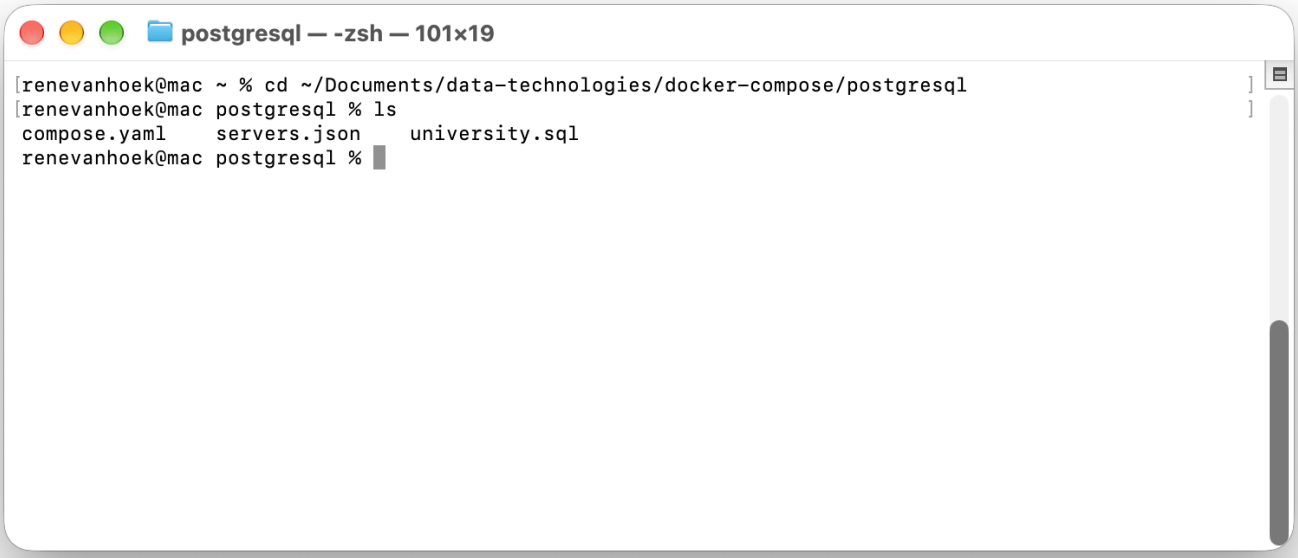
Check that the files are present

Run the following command:

```
ls
```

You should see the following three files:

```
compose.yaml
servers.json
university.sql
```



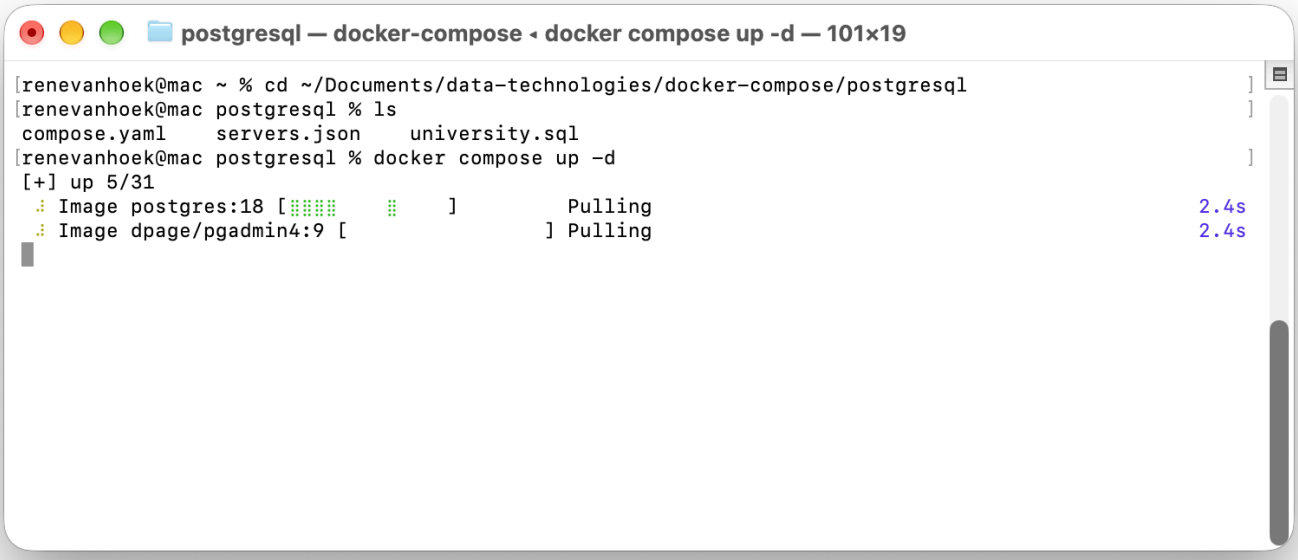
Check: If you do not see these three files, do not continue. Check that you have opened the correct directory.

Start the installation

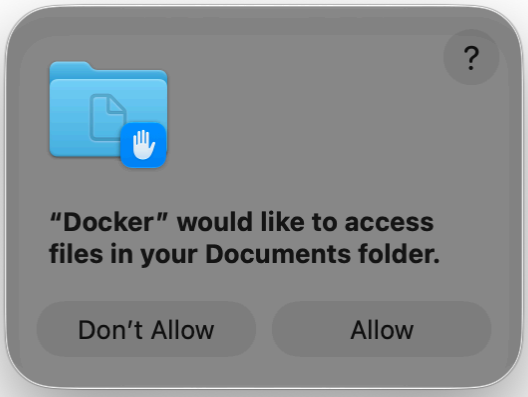
Run the following command:

```
docker compose up -d
```

Docker will now automatically download and install PostgreSQL and pgAdmin. This may take a few minutes the first time.



> On macOS, you may be asked whether Docker can access files in the Documents folder. Click ‘Allow’.



Check that the installation was successful

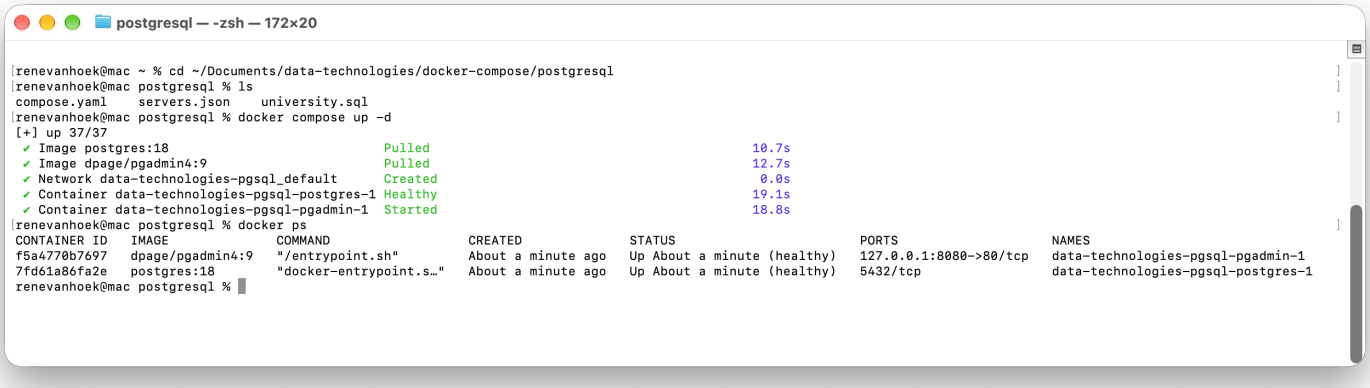
Run the following command:

```
docker ps
```

You should see two containers:

- postgres
- pgadmin

Both containers should have the status `Up`.

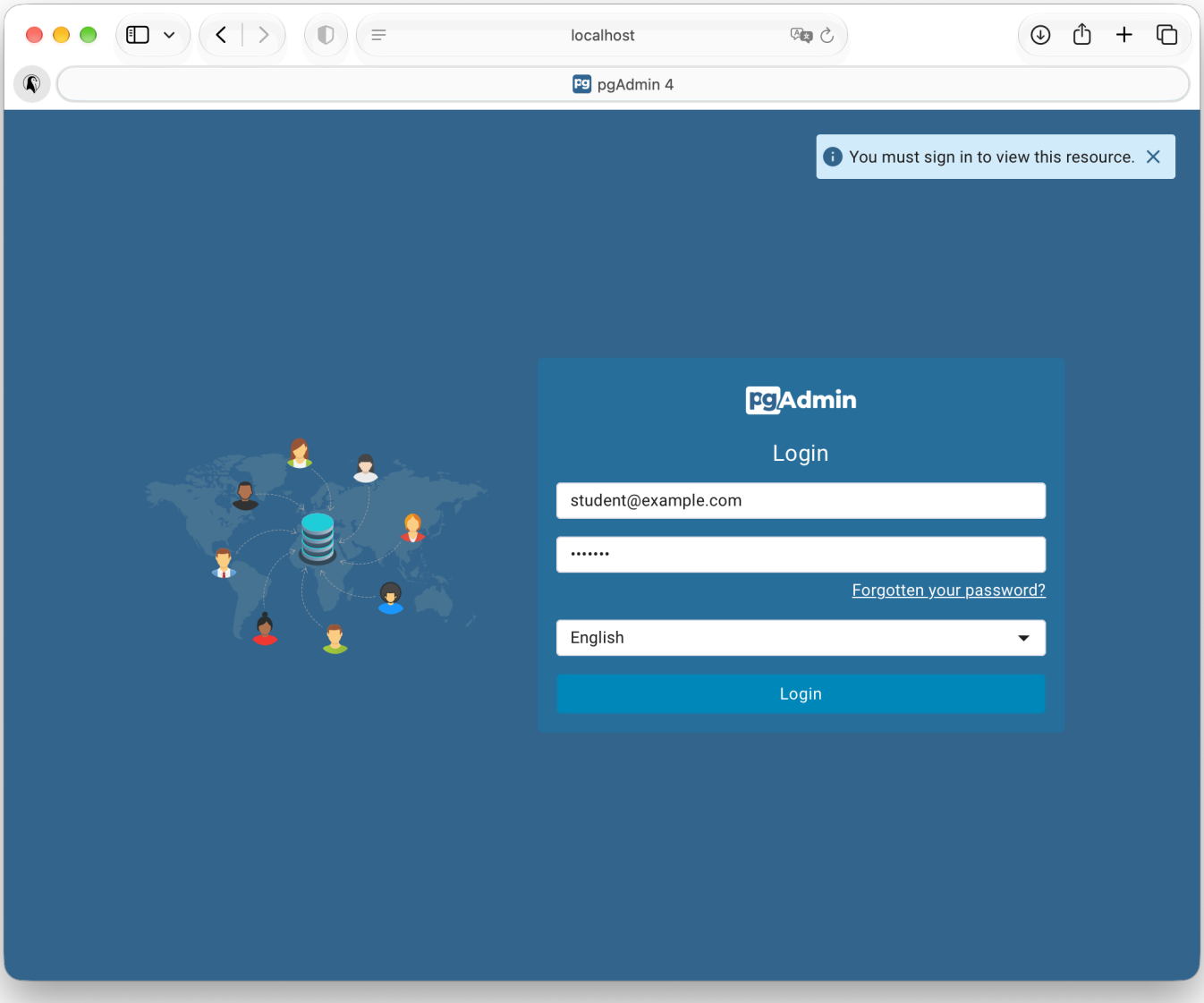


Step 6 - Test PostgreSQL using pgAdmin

We will check that PostgreSQL is installed correctly and that the `university` database is available. We will use pgAdmin to do this.

Log in to PgAdmin

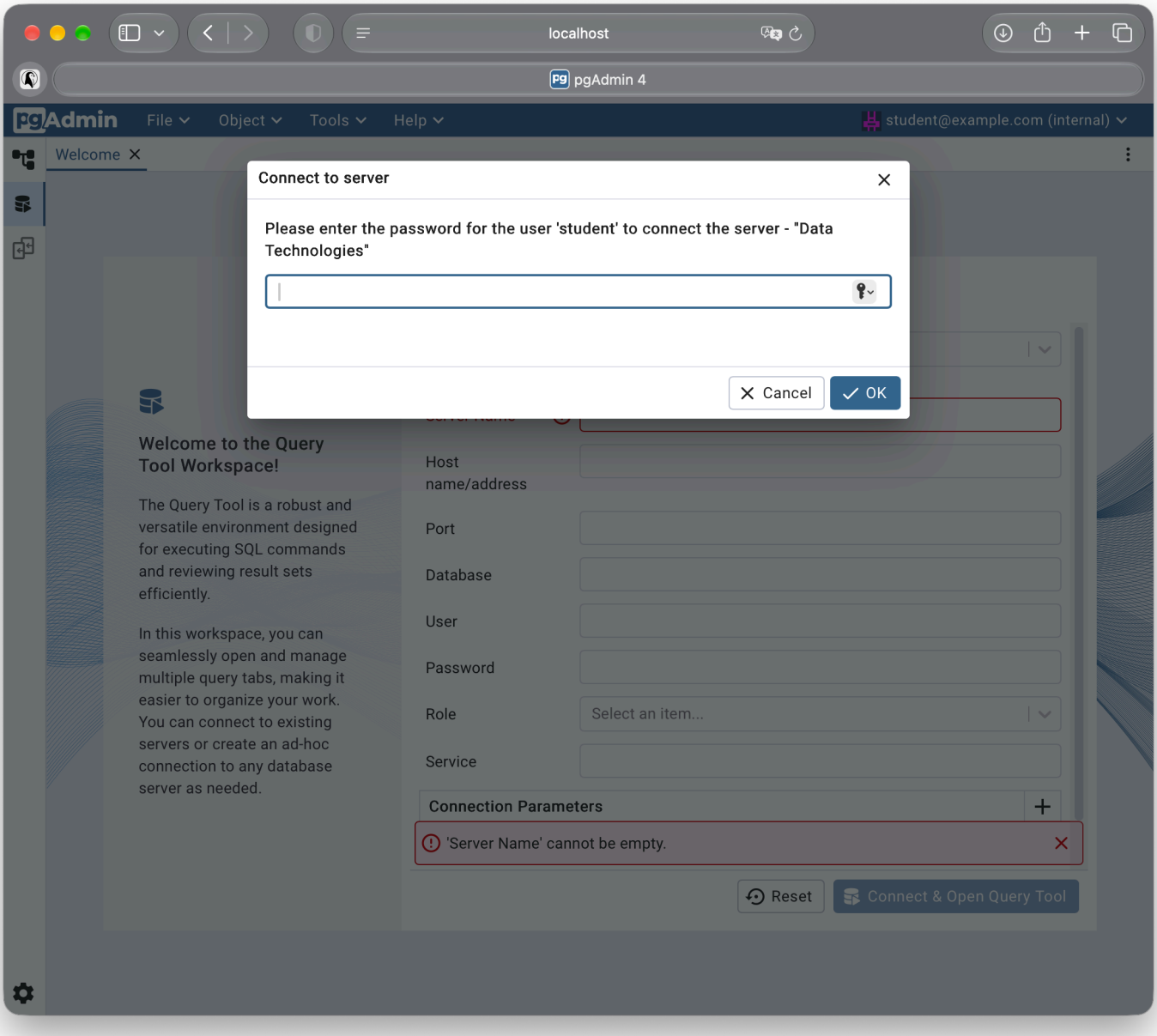
- Open your web browser (Edge, Safari or Chrome)
- Go to: http://localhost:8080
- You will see the pgAdmin login screen. Enter the following details:
    - Email address: `student@example.com`
    - Password: `student`
- Click Login

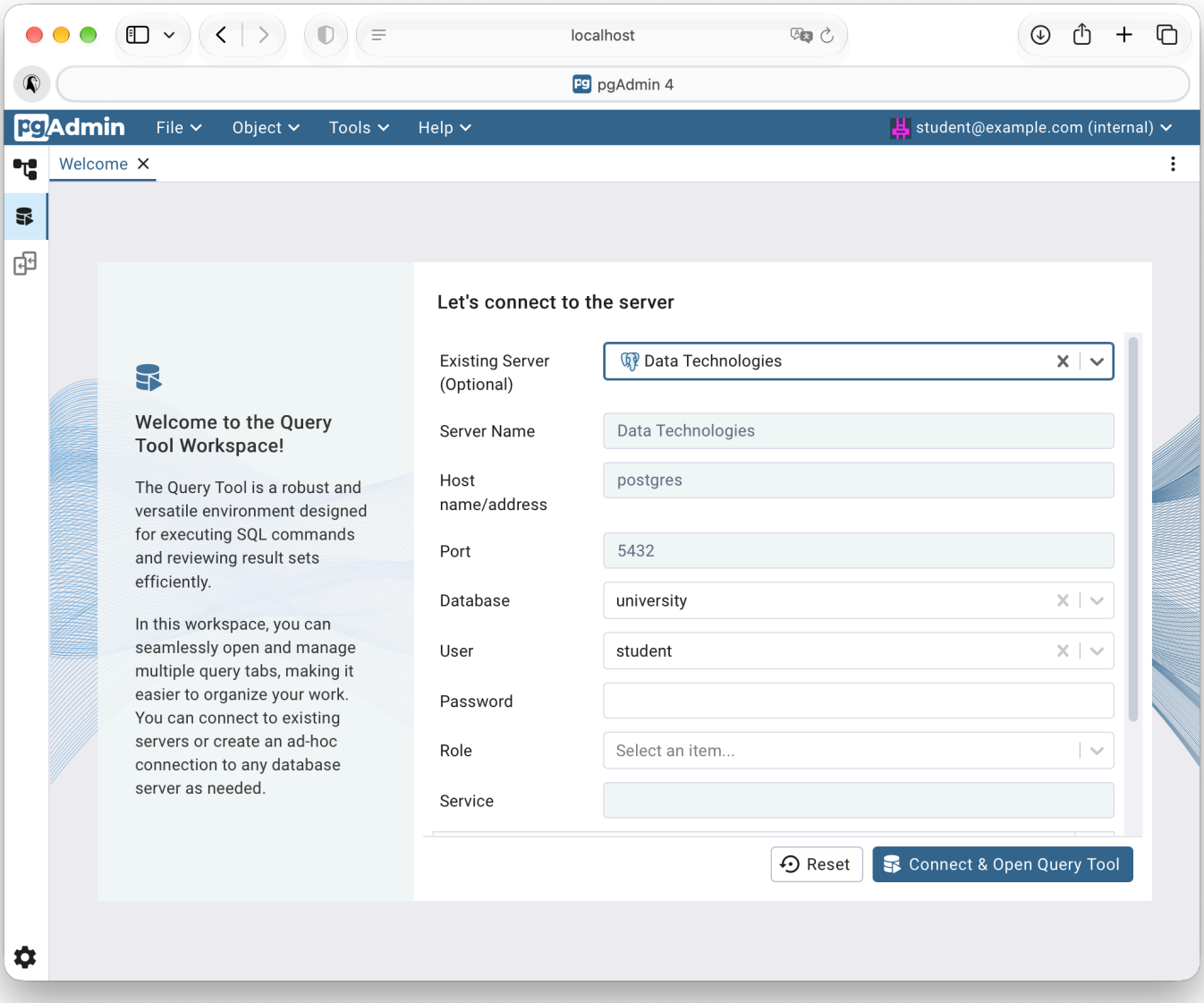


Open the Query Tool

The Query Tool lets you run SQL queries on the database.

- Click the **Query Tool icon** in the left sidebar (the middle of the three icons).
- The Query Tool welcome screen appears.
- Under **Existing Server**, select the `Data Technologies` server.
- If the **Connect to server** window appears:
  - Enter the password `student`.
- Check that **Database** is set to `university`.
- Enter the password `student` under **Password** if prompted.
- Click **Connect & Open Query Tool**.


localhost
pgAdmin 4
pgAdmin
File
Object
Tools
Help
student@example.com (internal)
Welcome
Connect to server
Please enter the password for the user 'student' to connect the server - "Data Technologies"
Cancel
OK
Welcome to the Query Tool Workspace!
The Query Tool is a robust and versatile environment designed for executing SQL commands and reviewing result sets efficiently.
In this workspace, you can seamlessly open and manage multiple query tabs, making it easier to organize your work. You can connect to existing servers or create an ad-hoc connection to any database server as needed.
Host name/address
Port
Database
User
Password
Role
Select an item...
Service
Connection Parameters
'Server Name' cannot be empty.
Reset
Connect & Open Query Tool



Run your first SQL query

The Query Tool is now open. We will retrieve the first twenty students from the database.

- Click the large white input field under the Query tab.
- Type the following SQL query:

```
SELECT * FROM students LIMIT 20;
```

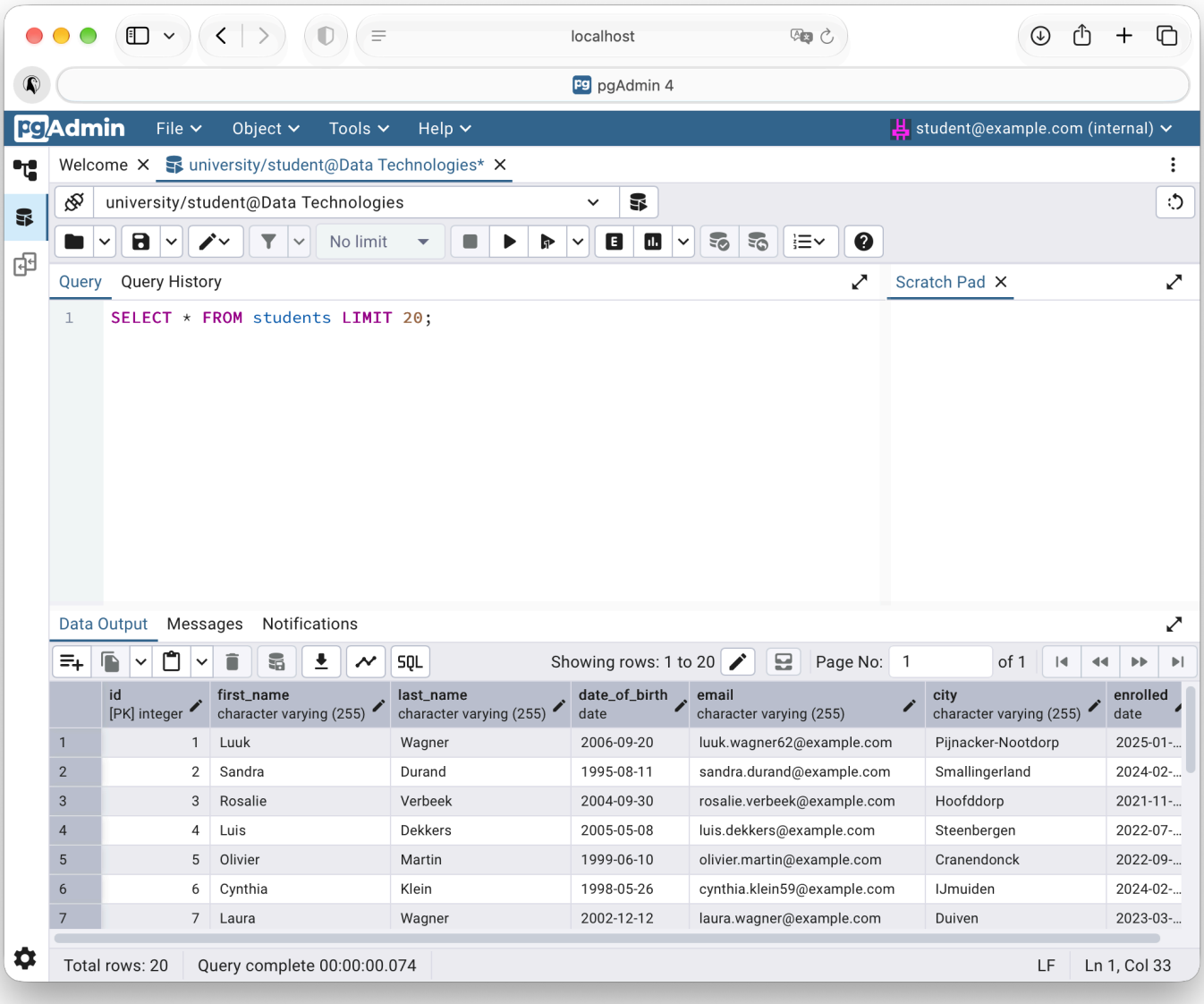
- Click the **Execute script** button (the triangular play icon ▶) in the toolbar.
- The query results appear at the bottom of the screen.

Check the result

If everything worked correctly, you will see a table with twenty students under Data Output. The table includes the following columns: `id`, `first_name`, `last_name`, `date_of_birth`, `email` and `city`.

If you can see twenty students, the installation was successful!

PostgreSQL is running on your laptop, pgAdmin is connected to the database, and you can run SQL queries.



Docker – Useful commands

Use the commands below to stop, restart or completely remove PostgreSQL and pgAdmin.

> Note: run these commands in the folder containing the `compose.yaml` file:
> `Documents → data-technologies → docker-compose → postgresql`.
>
> Use Terminal (macOS) or PowerShell (Windows).

Stop everything

```
docker compose down
```

This command stops and removes the PostgreSQL and pgAdmin containers.

Your data is preserved. The database and pgAdmin settings are stored in Docker volumes and are not deleted.

Start everything

```
docker compose up -d
```

This command starts PostgreSQL and pgAdmin.

If the containers do not yet exist, Docker creates them automatically. Previously saved data remains available.

Remove everything

```
docker compose down -v
```

This command stops the containers and also removes their associated Docker volumes.

Warning: this deletes all saved data! This applies to both the PostgreSQL database and the pgAdmin settings.

If you then run `docker compose up -d` again, the `university` database is recreated and populated with the data from `university.sql`.